

Experiment No: 03

Date: 17/09/2026

Title: Bank Account Management System using Inheritance and Method Overriding

Objective / Aim

To design and implement a Bank Account Management System in Java using classes, objects, constructors, inheritance, method overriding, and static methods, demonstrating deposit, withdrawal, interest calculation, overdraft facility, and fund transfer operations.

System Requirements

Hardware

- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage

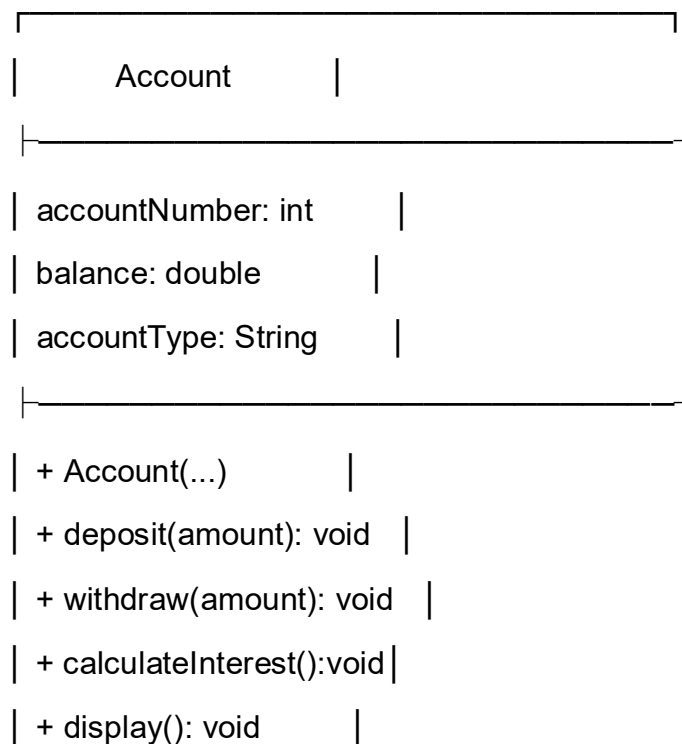
Software

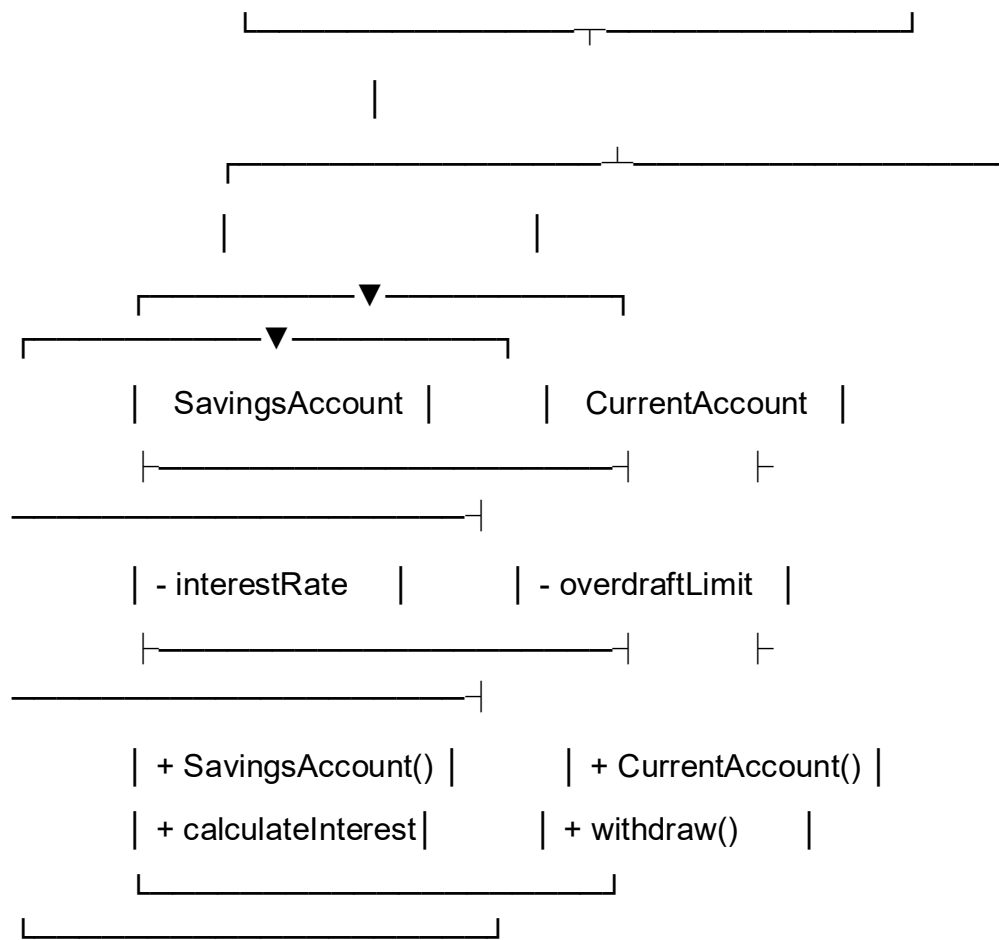
- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK 17 or JDK 21)
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt / Terminal

Algorithm / Logic Description

1. Create an "Account" class with account number, balance, and account type.
2. Define methods for deposit, withdrawal, interest calculation, and displaying details.
3. Create "SavingsAccount" by inheriting "Account" and override the interest calculation method.
4. Create "CurrentAccount" by inheriting "Account" and override withdrawal to support overdraft.
5. Create a "transfer()" method to transfer money between two accounts.
6. Create Savings and Current account objects.
7. Perform deposit, interest, withdrawal, and transfer operations.
8. Display the final account details.

UML Class Diagram





Relationship: "SavingsAccount" and "CurrentAccount" inherit from the "Account" class. "SavingsAccount" overrides "calculateInterest()", while "CurrentAccount" overrides "withdraw()".

Source Code

```
package javacore;

// Account class represents a general bank account
class Account {

    // Data members of the Account class
    int accountNumber;
    double balance;
    String accountType;

    // Parameterized constructor
    Account(int accountNumber, double balance, String accountType) {
        this.accountNumber = accountNumber;
        this.balance = balance;
        this.accountType = accountType;
    }

    // Method to deposit money
    void deposit(double amount) {

        // Check whether amount is positive
        if (amount > 0)
            balance += amount;
    }

    // Method to withdraw money
    void withdraw(double amount) {
```

```
// Check balance before withdrawal
if (amount > 0 && amount <= balance)
    balance -= amount;
else
    System.out.println("Insufficient balance");
}
```

```
// Method to calculate interest
void calculateInterest() {
    System.out.println("No interest");
}
```

```
// Method to display account details
void display() {

    System.out.println("Account Number: " + accountNumber);
    System.out.println("Account type: " + accountType);
    System.out.println("Balance: " + balance);
}
}
```

```
// SavingsAccount class inherits Account
class SavingsAccount extends Account {

    // Interest rate
    private double interestRate;
```

```
// Constructor
SavingsAccount(int no, double balance, double rate) {

    // Calling parent constructor
    super(no, balance, "Savings Account");

    interestRate = rate;
}

// Overriding calculateInterest()
@Override
void calculateInterest() {

    // Calculate interest
    double interest = balance * interestRate / 100;

    // Add interest to balance
    balance += interest;

    System.out.println("Interest: " + interest);
}
}

// CurrentAccount class inherits Account
class CurrentAccount extends Account {

    // Overdraft limit
    private double overdraftLimit;
```

```
// Constructor
CurrentAccount(int no, double balance, double limit) {

    // Calling parent constructor
    super(no, balance, "Current Account");

    overdraftLimit = limit;
}

// Overriding withdraw() for overdraft
@Override
void withdraw(double amount) {

    // Check overdraft limit
    if (amount > 0 && amount <= balance + overdraftLimit)
        balance -= amount;
    else
        System.out.println("Exceeds overdraft limit");
}
}

// Main class
public class BankAccountManagement {

    // Static method to transfer money
    static void transfer(Account from, Account to, double amount) {
```

```
// Check amount and balance
if (amount > 0 && amount <= from.balance) {

    // Deduct from source account
    from.balance -= amount;

    // Add to destination account
    to.balance += amount;

    System.out.println("Transfer successful");

} else {
    System.out.println("Transfer failed");
}
}
```

```
// Main method
public static void main(String[] args) {

    // Creating SavingsAccount object
    SavingsAccount savings =
        new SavingsAccount(2002, 500000, 3);

    // Creating CurrentAccount object
    CurrentAccount current =
        new CurrentAccount(2003, 5000, 3000);

    // Display savings account
```

```
savings.display();

// Deposit money
savings.deposit(3000);

// Calculate interest
savings.calculateInterest();

// Display current account
current.display();

// Withdraw using overdraft
current.withdraw(6000);

// Transfer money
transfer(savings, current, 2000);

// Display final savings details
savings.display();

// Display final current details
current.display();
}
}
```

Compilation & Execution Commands

Save the source code as:

BankAccountManagement.java

If you are using the "package javacore;" statement, save the file inside the appropriate "javacore" package/folder.

Compile the program:

```
javac BankAccountManagement.java
```

Run the program:

```
java BankAccountManagement
```



Output Screen

Account Number: 2002

Account type: Savings Account

Balance: 500000.0

Interest: 150900.0

Account Number: 2003

Account type: Current Account

Balance: 5000.0

Transfer successful

Account Number: 2002

Account type: Savings Account

Balance: 651900.0

Account Number: 2003

Account type: Current Account

Balance: 3000.0

Result

The Bank Account Management System was successfully implemented in Java using classes, objects, constructors, inheritance, method overriding, and static methods. The program successfully performs deposit, withdrawal, interest calculation, overdraft, and fund transfer operations.